NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

MIL-PRF-6106/10D(USAF)
19 December 1989
SUPERSEDING
MIL-R-6106/10C(USAF)
17 MARCH 1986

PERFORMANCE SPECIFICATION SHEET

RELAY, ELECTROMAGNETIC, NORMALLY OPEN, CONTINUOUS DUTY, 60 AMP, 3PST, ENVIRONMENTALLY SEALED

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relay described herein shall consist of this specification and the latest issue of MIL-PRF-6106.

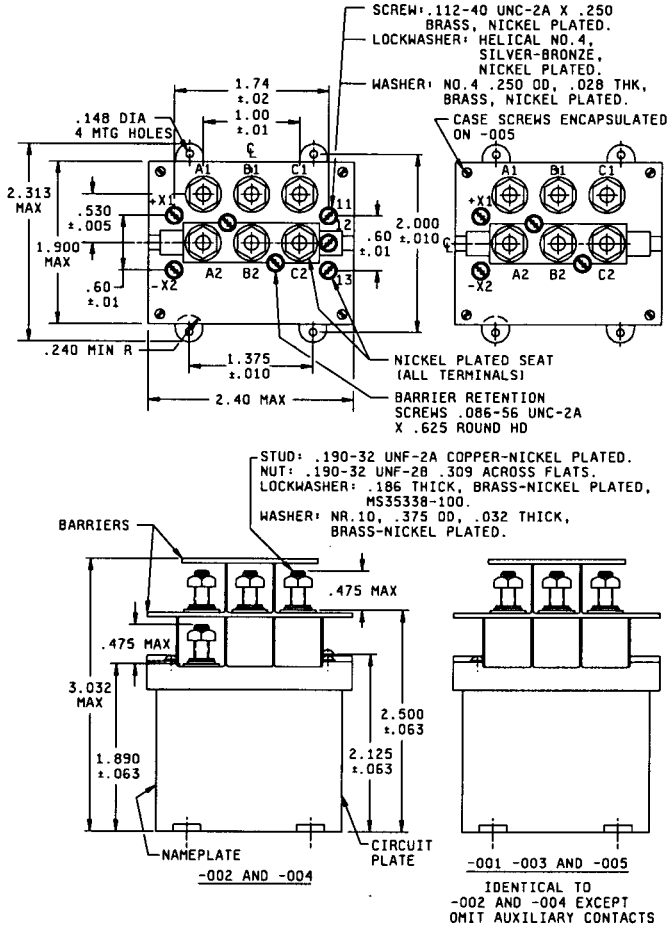
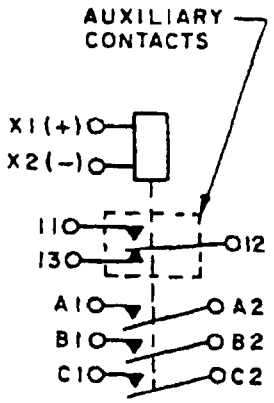


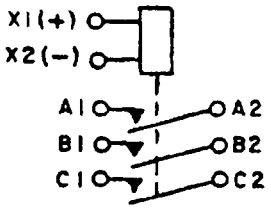
FIGURE 1. Relay, outline drawing.

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

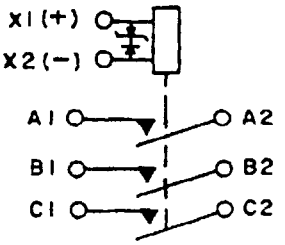
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CIRCUIT
-002 & -004



CIRCUIT
-001 & -003



CIRCUIT
-005 7/ 8/

INCHES	mm	INCHES	mm	INCHES	mm
.005	0.13	.250	6.35	1.375	34.92
.010	0.25	.309	7.85	1.74	44.2
.02	0.5	.375	9.52	1.890	48.00
.028	0.71	.440	11.18	1.900	48.26
.032	0.81	.530	13.46	2.000	50.80
.063	1.60	.600	15.24	2.115	53.72
.148	3.76	.625	15.88	2.313	58.75
.186	4.72	1.00	25.4	2.40	61.0
.240	6.10	1.060	26.92	2.500	63.50
				3.032	77.01

NOTES:
1. Dimensions are in inches
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ± 031 (0.79 mm).
4. Weights include cover, barriers, and terminal hardware.
5. When using relay at rated 60 amps, the use of lugs per Burndy part number YAEV6C-G8 or equal is recommended
6. Terminal temperature rise under continuous current conditions is 95°C maximum.
7. Semiconductor devices shall be JANTX or equivalent. Diodes shall have a peak inverse voltage of 800 V dc minimum.
8. -005 relays shall not operate or be damaged by reversed polarity

FIGURE 1. Relay, outline drawing - Continued.

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(D) TABLE I. Dash number erating characteristics

Dash no	Type	Coil data: Coil	Nominal: Volts	Nominal: Freq	Nominal: Res	Maximum: Volts	Maximum: Ampere: Inrush	Maximum: Ampere: Steady state	Max pickup voltage 1/: Normal 1/	Max pickup voltage 1/: High Temp Test	Max pickup voltage 1/: Cont Current Test	Hold voltage 1/	Drop out voltage 1/	Time milliseconds - maximum: Operate 2/	Time milliseconds - maximum: Release 3/	Time milliseconds - maximum: Bounce: Main: NO	Bounce: Main: NC	Bounce: Auxiliary: NO	Bounce: Auxiliary: NC
-001	I	X1, X2	28	DC	-	29	1 25	25	17	20	21.5	7 0	1 5	25	25	2	-	-	-
-002	I	X1, X2	28	DC	-	29	1 25	25	17	20	21.5	7 0	1 5	25	25	2	-	2	2
-003	I	X1, X2	115	400	-	124	4	1	85	95	100	40	10	25	35	2	-	-	-
-004	I	X1, X2	115	400	-	124	4	1	85	95	100	40	10	25	35	2	-	2	2
-005 4/	I	X1, X2	28	DC	-	29	1 25	25	17	20	21 5	7 0	1 5	25	15.	-	-	-	-

1/ Over the temperature range. 2/ With nominal coil voltage 3/ From nominal coil voltage

4/ -005 is same as -001 except for transient suppressed coil

TABLE II Rated contact load (amperes per pole) (case grounded)

Type of load	Life operating cycles x 10^3	28 V dc 1/: Main: NO	28 V dc 1/: Main: NC	28 V dc 1/: Auxiliary: NO	28 V dc 1/: Auxiliary: NC	115 V ac, 1 phase 1/: Main: 400 Hz	115 V ac, 1 phase 1/: Main: 60 Hz	115 V ac, 1 phase 1/: Auxiliary: 400 Hz	115 V ac, 1 phase 1/: Auxiliary: 60 Hz	115/200 V ac, 3 phase 1/: Main: 400 Hz	115/200 V ac, 3 phase 1/: Main: 60 Hz	115/200 V ac, 3 phase 1/: Auxiliary: 400 Hz	115/200 V ac, 3 phase 1/: Auxiliary: 60 Hz
Resistive	100		-	3	3	60	-	3	-	60	-	-	-
Inductive	100	-	-	1 5	1 5	60	-	1 5	-	60	-	-	-
Motor	100	-	-	-	-	40	-	-	-	40	-	-	-
Lamp		-	-	5	5	20	-	5	-	20	-	-	-
Transfer load 2/													
Mechanical life reduced current	200	-	-	75	75	15	-	75	-	15	-	-	-
Minimum 3/ current	50	-	-	Per spec		4	-	-	-	4	-	-	-

1/ Absence of value indicates relay is not rated for that application.

2/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplies at rating

3/ Minimum current to be conducted at +71°C Number of minimum current cycles following continuous current test - 5000

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REQUIREMENTS:

Weight: -001, -002, and -005 10 ounces maximum.
-003 and 004; 11 ounces maximum.

Temperature range -65°C to +71°C.

Maximum altitude rating: 50,000 feet 1/.

Shock g-level: 1/2 sine, 25 g's.

Duration. 6 to 9 ms.

Maximum duration contact opening 1 ms.

Vibration-sinusoidal: See table III.

TABLE III. Vibration-sinusoidal.

Dash No.	Vibration level					
	5-10 Hz	10-55 Hz	55-400 Hz	400-800 Hz	800 - 2000 Hz	
					-55°C and 25°C	+71°C
-001	.08 DA	.06 DA	10 g's	10 g's	10 g's	7 g's
-002			10 g's	8 g's	8 g's	7 g's
-003			10 g's	10 g's	10 g's	7 g's
-004			10 g's	8 g's	8 g's	7 g's
-005			10 g's	10 g's	10 g's	7 g's

Vibration (random) Not applicable.

Acceleration: 15 g's.

Seal

Applicable specification MIL-STD-202, method 112, test condition C, procedure 1V.

Leak rate 6×10^{-4} atm cm^3/s 2/.

Terminal strength:

Applicable specification MIL-STD-202, method 211, test condition E.

Torque. 24 pound-inches.

Insulation resistance, initial. 200 megohms minimum.

After life or enviromental tests: 100 megohms minimum.

1/ Application range can be extended to 80,000 feet by suitably encapsulating the terminals.
2/ During the test, pressure shall be reduced below 10 microns for a minimum of 10 seconds.

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Dielectric withstanding voltage (sea level)· 2 to 5 seconds.

	Initial		After life tests	
	28 V dc	115 V ac	28 V dc	115 V ac
Coil to case- - - - - - - - - - - - - -	1250	1500	1000	1125
Auxiliary contacts- - - - - - - - - - -	1250	1250	1000	1000
Across open power contacts- - - - - - -	---	1250	---	625
All other points- - - - - - - - - - - -	1800	1800	1350	1350

Dielectric withstanding voltage (altitude). 1 minute.

	28 V dc	115 V ac
Coil to case- - - - - - - - - - - - - - -	500	500
Auxiliary contacts- - - - - - - - - - - -	500	500
All other points- - - - - - - - - - - - -	700	700

Maximum contact drop, initial: .150 volt.

After life test: .200 volt.

Overload current 3/· 320 amperes.

Rupture current· 400 amperes.

Coil suppression (transient voltage, back emf) 42 V dc maximum.

Duty rating: Continuous.

RFI specification (applicable to coil circuits of ac operated relays): MIL-STD-461 and MIL-STD-462.

Part number. M6106/10- (dash number from table I)

TABLE IV. Qualification by similarity.

	Loads						Dynamics 1/			Environmental 2/		
	Type I			Type I ER								
Dash number	A	B	C	D	E	F	A	X	Y	A	B	C
-001	4						2			4		
-002	4 2/						2			4		
-003		1						1			1	
-004		1 2/						1			1	
-005	1								1			1

1/ All units must be tested, reference the appendix of MIL-R-6106, construction, internal.

2/ Test unit with auxiliary contacts, reference the appendix of MIL-R-6106, construction, internal.

3/ Number of rupture cycles after minimum current test - 5.

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Custodian
Air Force - 85

Review activities.
Air Force - 11, 99
DLA - ES

Preparing activity
Air Force - 85

Agent
DLA - ES

(Project 5945-F703-02)

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